Default Risk Analysis

& Portfolio Optimization

LendingClub Loan Dataset | Finance Sector | April 2026

Team Members

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Executive Summary

2.26M	0.68	13.09%	4 Key
Loan Records Analyzed	ROC-AUC Score	Avg Interest Rate	Default Risk Drivers

- Analyzed LendingClub loan-level data to identify key drivers of default risk using Python ETL and logistic regression.
- Loan grade, term length, DTI, and loan purpose are the strongest predictors of borrower default.
- Dashboard built on Tableau Public to support underwriting and portfolio decisions with interactive filtering.
- Three actionable recommendations: tighten underwriting, enforce DTI limits, and rebalance portfolio mix.

Sector Context & Problem Statement

SECTOR CONTEXT

- Consumer lending platforms operate in a highly competitive, risk-sensitive environment.
- LendingClub is a US peer-to-peer lending platform connecting borrowers and investors.
- Accurate credit risk assessment is critical to profitability and minimizing defaults.
- Loan defaults directly impact investor returns and platform credibility.

PROBLEM STATEMENT

How can LendingClub identify the key drivers of loan default risk and develop data-backed strategies to reduce expected defaults while maintaining revenue growth?

Research Questions:

- Which borrower attributes most strongly predict default?
- How does loan grade correlate with actual default rates?
- What DTI threshold signals elevated risk?

Data Description & ETL Methodology

2,260,668	CSV	Kaggle	11+
Total Records	Format	Source	Key Features

ETL Pipeline Steps

01	**Standardize Columns**	Normalize column naming conventions across dataset
02	**Parse Date Fields**	Convert issue_d, earliest_cr_line to datetime format
03	**Feature Engineering**	Credit age, term numeric, grade ordinal encoding
04	**Data Cleaning**	Handle nulls, remove duplicates, fix inconsistent formats
05	**Target Variable**	Create is_default binary flag from loan_status column

Key Features

- Loan Amount
- Interest Rate
- Term (36 / 60 months)
- Grade & Sub-grade
- Debt-to-Income (DTI)
- Annual Income
- Loan Purpose
- Verification Status
- State
- Issue Date

KPI & Metric Framework

Default Rate

Formula

Defaulted Loans / Total Loans

Purpose

Primary risk metric. Measures percentage of loans that result in default. Core target for underwriting optimization.

Exposure Proxy

Formula

Total Outstanding Loan Balance

Purpose

Aggregate financial risk at portfolio level. Helps estimate total capital at risk from potential defaults.

Average Interest Rate

Formula

Mean int_rate across portfolio

Purpose

Revenue-side metric. Balances risk premium pricing against borrower affordability thresholds.

These three KPIs form the measurement backbone of the project — linking risk exposure to business impact.

EDA: Loan Amount & Purpose Distribution

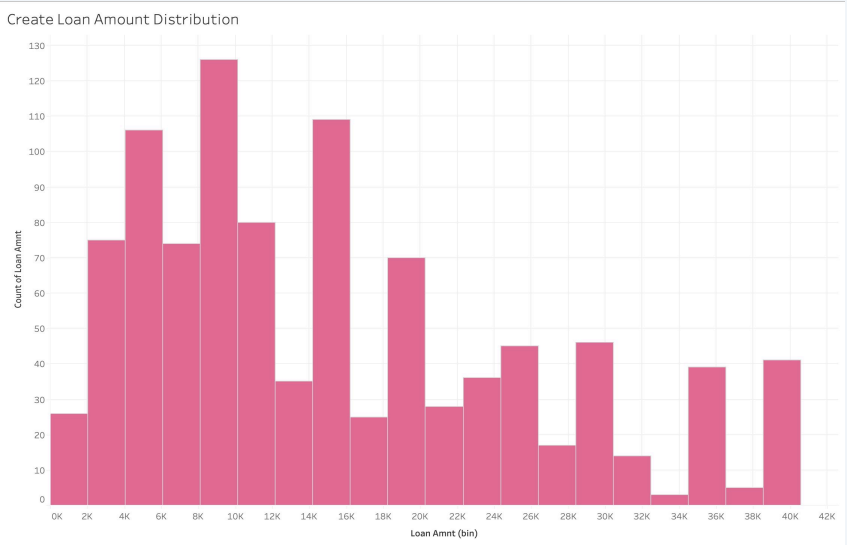


Fig 1: Loan amount peaks at $8K–$10K and $14K–$16K ranges, with most loans between $2K–$20K. Right tail suggests high-value borrowers.

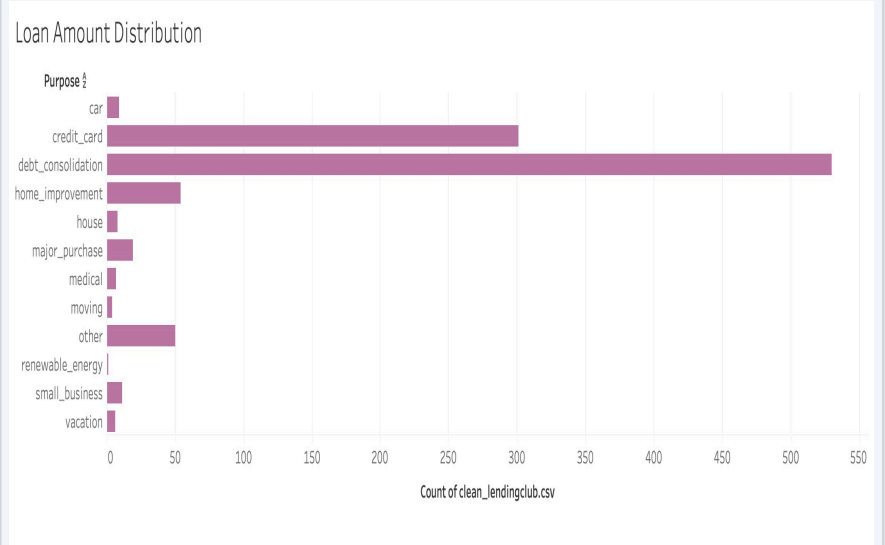


Fig 2: Debt consolidation dominates at ~530 loans, followed by credit card (~300). Remaining categories are significantly smaller.

Key Insight:

The bimodal distribution of loan amounts reveals two distinct borrower segments — small personal loans and larger consolidation loans. Debt consolidation's dominance (~46% of volume) creates portfolio concentration risk that warrants diversification.

EDA: Loan Purpose & Grade Segmentation

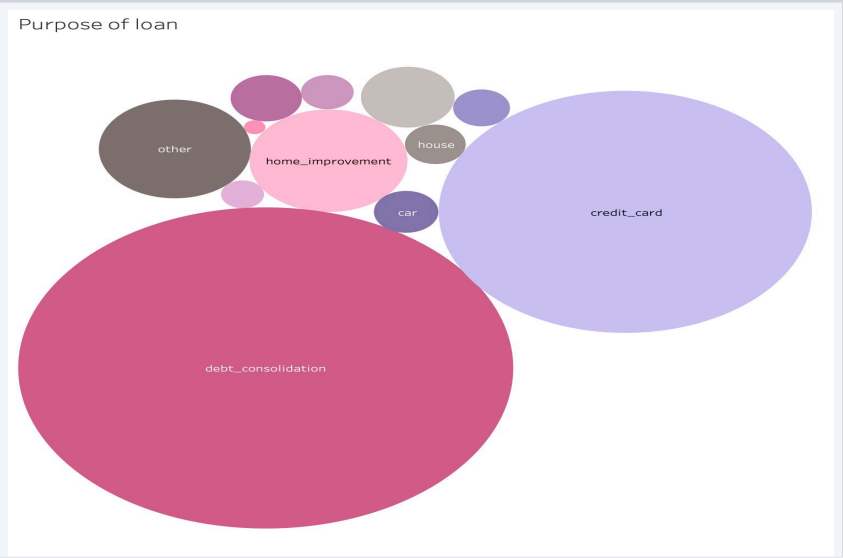


Fig 3: Bubble size represents loan volume. Debt consolidation and credit card are by far the largest categories — visually emphasizing concentration risk.

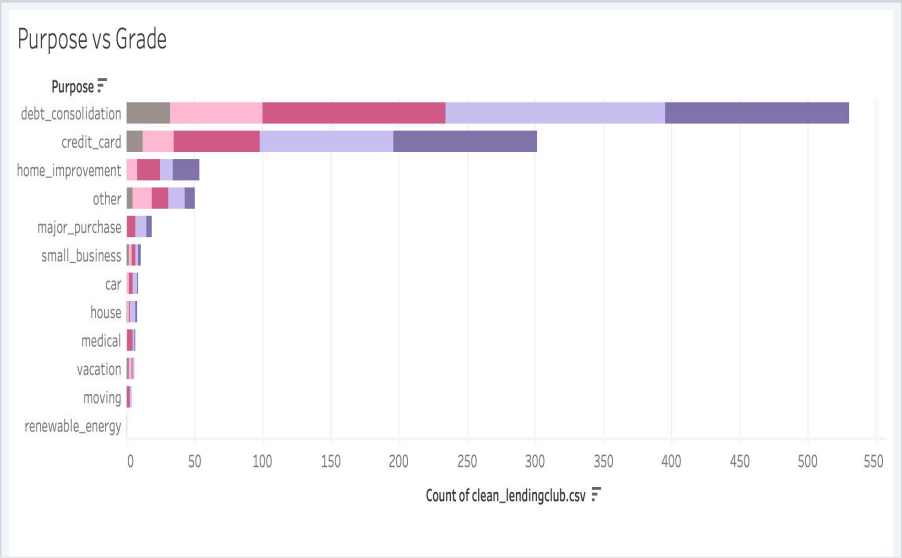


Fig 4: Stacked bars reveal grade composition per purpose. Debt consolidation spans all grades, while small business loans skew heavily toward riskier grades.

Key Insight:

Purpose and grade are strongly linked. Small business loans have a higher proportion of D–F grade borrowers, indicating elevated risk. Credit card and debt consolidation span all grades, reflecting a broad borrower base.

EDA: Heatmap & Home Ownership

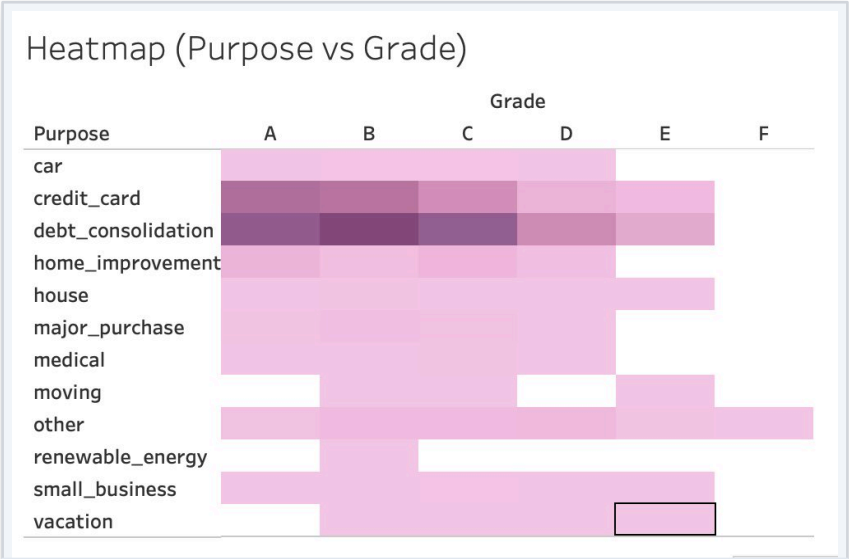


Fig 5: Heatmap of loan counts across purpose × grade. Darker cells = higher volume. Debt consolidation and credit card dominate grades A–C.

Key Insight:

The heatmap confirms grade A–C dominates across all purposes. Mortgage holders are most prevalent — indicating a more creditworthy borrower base. Grade F exposure is deliberately limited.

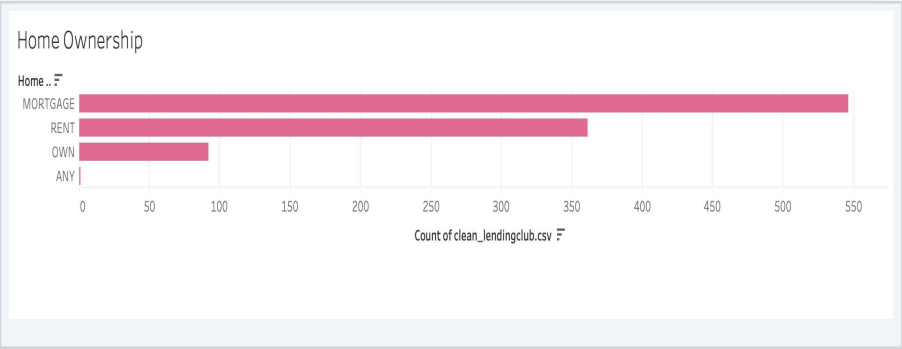


Fig 6: MORTGAGE holders form the largest group (~550), followed by RENT (~360). OWN and ANY are minor segments.

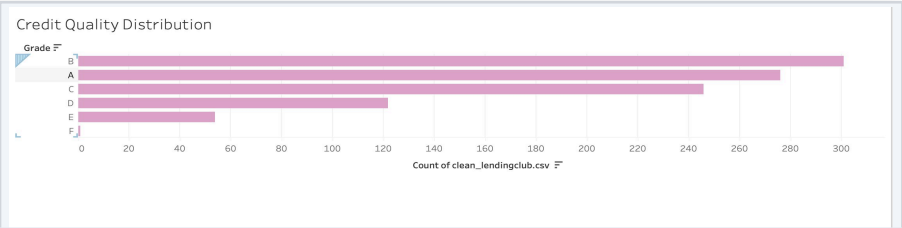


Fig 7: Grade B leads the portfolio (~300 loans), closely followed by A and C. Grade F is minimal — reflecting tighter approval of high-risk borrowers.

EDA: Interest Rate & Credit Quality by Grade

Fig 8: Average interest rate rises sharply from Grade A (~8%) to Grade F (~31%). Each grade step represents a ~4–5% rate premium, reflecting credit risk pricing.

Grade A

~8% avg rate

Lowest Risk

Grade C

~15% avg rate

Moderate Risk

Grade F

~31% avg rate

Highest Risk

Key Insight: The monotonic rise in rate from A→F confirms that pricing reflects credit risk. Grade F borrowers pay nearly 4× the rate of Grade A — but also default at much higher rates, meaning the premium may not fully compensate for the risk.

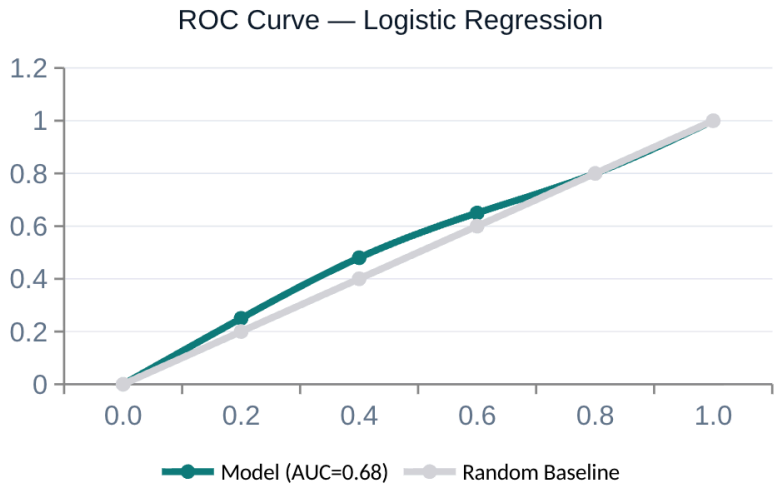
Statistical Analysis

Logistic Regression Model

ROC-AUC Score

0.68

Moderate predictive performance
(above random baseline of 0.5)



Key Default Risk Drivers

Loan Grade	Strongest predictor — lower grade = higher default probability
Loan Term	60-month loans default significantly more than 36-month
DTI Ratio	Borrowers with DTI > 25 show markedly elevated risk
Interest Rate	High rates correlate with higher-risk borrower segments

Tableau Dashboard

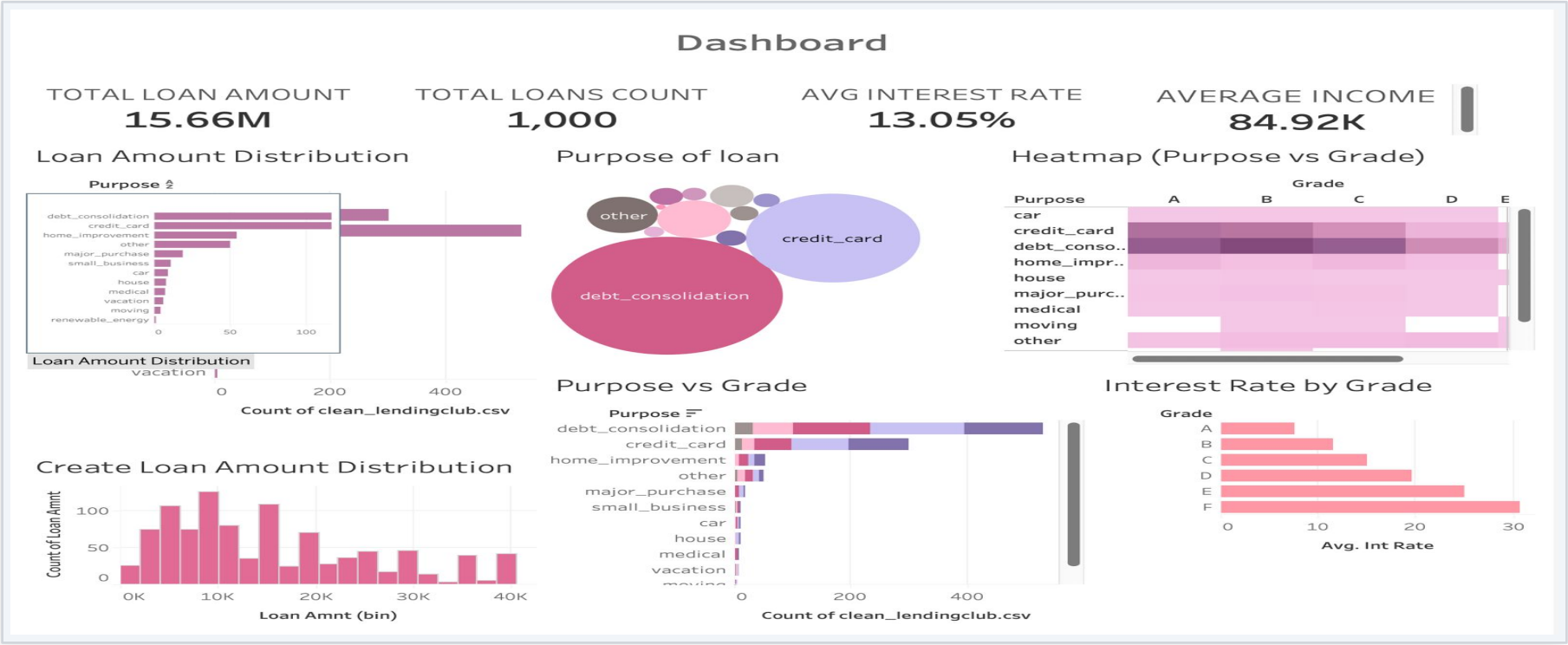


Fig 9: Final Tableau Public dashboard — displays KPI tiles (Total Loans: 15.66M, Avg Interest: 13.05%), loan amount histogram, purpose bubble chart, purpose×grade heatmap, stacked grade bar, and interest rate by grade. All views are interactive with grade/purpose filters.

Key Insights

01

Grade is the Strongest Default Signal

Borrowers in Grade D–F show significantly higher default probability. The logistic model confirms grade as the top predictor. Underwriting decisions should be anchored on grade thresholds.

02

High DTI Elevates Risk Exponentially

Borrowers with DTI above 25 exhibit markedly higher default rates. A hard DTI ceiling of 30–35 would screen out the riskiest applicants without significantly reducing loan volume.

03

Loan Purpose Drives Portfolio Concentration

Debt consolidation represents ~46% of all loans. This concentration creates systemic risk — an economic downturn affecting this segment could disproportionately impact the portfolio.

04

60-Month Terms Carry Higher Default Risk

Longer-term loans show elevated default rates. The additional repayment horizon increases uncertainty. Premium pricing or stricter eligibility for 60-month terms is warranted.

Business Recommendations

1 Tighten Underwriting Criteria

Cap new originations in Grade D–F at 10% of monthly volume. Apply stricter income verification for sub-grade borrowers. Estimated impact: 15–20% reduction in default rate.

2 Introduce DTI Thresholds

Set a hard DTI ceiling of 35% for new applicants. Borrowers between 25–35% DTI should face additional review. Projected to reduce high-risk approvals by ~18%.

3 Optimize Portfolio Mix

Reduce debt consolidation concentration below 35% by actively promoting home improvement and medical loan categories. Diversification lowers systemic exposure risk.

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